

Modern Hopfield Networks for SLR

This repository holds my work in using Modern Hopfield Networks to advance the task of Isolated Sign Language Recognition (ISLR) in the VLM Lab directed by Dr. Vassilis Athitsos. The following README is a short technical explanation of the methodology used.

Methodology

In our efforts to develop methodologies for an American Sign Language Dictionary, we reframe ISLR from a classification task into a dictionary-retrieval task. When humans recognize signs, we gather phonological features of the sign like hand-shape and compare against our learned bank of signs. Thus, we theorize that a similar process through **associative memory** can achieve significant improvements over an encode-and-classify baseline. For this task, we first encode using VideoMaeV2 [1]. These encodings are then passed into a Modern Hopfield Network [2], specifically the HopfieldLayer. We note that the exponential storage capacity of Modern Hopfield Networks [2] allow for the association of the large vocabulary of sign languages.

Transformations and Training Strategies

We follow the same training strategies outlined in [3], except for boundary shifts as the ASL Citizen dataset [4] does not annotate sign boundaries. As the authors of [3] note, some signs in ASL attribute their meaning to handedness. Thus, we remove horizontal flips as a training augmentation, unlike several previous works in the literature.

For our VideoMaeV2 encoder, we fine tune using LP-FT [5] from the MCG-NJU/videomae-base-finetuned-ssv2 weights on HuggingFace. We then use only the encoder, and discarding the linear probe, in our full model. When training the full model and the MLP classifier baseline, we freeze the weights on the encoder and train only the Hopfield Network or MLP. To aid training, we initialize the stored patterns of the Hopfield Layer using examples from the training set.

Results

We train and evaluate our methodology on the ASL Citizen [4] dataset.

Bibliography

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